GPU-Accelerated Cryptographic Operations for Pars Network

Pars Network Technical Whitepaper PNP-011

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Abstract

We present a GPU acceleration framework for the three computationally intensive cryptographic subsystems of the Pars Network: FHE bootstrapping for private messaging and governance (PNP-009), Number Theoretic Transforms for post-quantum lattice operations (PNP-004), and batch zero-knowledge proof generation for identity verification (PNP-006). All three subsystems share the NTT as a computational bottleneck. We design a unified GPU kernel library that serves all three workloads through a common polynomial arithmetic layer, achieving 50–90× speedup over single-core CPU across operations. We target both data-centre GPUs for validator nodes and consumer GPUs for diaspora users, and provide concrete benchmarks on NVIDIA Ampere and Ada architectures. The framework integrates with the Pars mesh network (PNP-002) to enable GPU-assisted computation for resource-constrained nodes via secure offloading.

Keywords: GPU acceleration, NTT, FHE bootstrapping, post-quantum cryptography, zero-knowledge proofs, batch proving, CUDA kernels

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1 Introduction

The Pars Network deploys three families of computationally expensive cryptographic primitives:

1. **FHE bootstrapping** (PNP-009): each TFHE gate bootstrap requires an NTT-based polynomial multiplication. Encrypted search over 1000 messages demands ∼255,000 bootstraps.
2. **Post-quantum key exchange and signatures** (PNP-004): ML-KEM and ML-DSA operate on polynomial rings $\mathbb{Z}_q[X]/(X^N + 1)$, with NTT as the core operation.
3. **Zero-knowledge proof generation** (PNP-006): Groth16 and Plonk provers require multi-scalar exponentiation (MSM) and NTT over large fields.

All three workloads are embarrassingly parallel and dominated by the same algebraic primitive: the Number Theoretic Transform. A unified GPU library amortises development, testing, and deployment cost.

1.1 Design Principles

1. **Unified NTT kernel.** One optimised NTT implementation serves FHE, post-quantum, and ZK workloads via parameterisation over modulus, degree, and field.
2. **Batch-first.** All APIs accept batches of independent operations to maximise GPU occupancy.
3. **Consumer-grade targets.** Kernels must run on GPUs available to diaspora users (RTX 3060 and above), not only data-centre hardware.
4. **Constant-time.** All kernels execute in constant time to prevent timing side-channels.

2 Number Theoretic Transform

2.1 Definition

Definition 2.1 (NTT)**.** *The Number Theoretic Transform of a polynomial $a(X) = \sum_{i=0}^{N-1} a_i X^i$ over $\mathbb{Z}_q$ evaluates a at the N-th roots of unity $\omega^0, \omega^1, \ldots, \omega^{N-1}$ where ω is a primitive N-th root of unity modulo q:*

$$\hat{a}_j = \sum_{i=0}^{N-1} a_i \cdot \omega^{ij} \bmod q$$

The NTT has $O(N \log N)$ complexity via the Cooley–Tukey butterfly decomposition, identical in structure to the FFT but over $\mathbb{Z}_q$ rather than $\mathbb{C}$.

2.2 GPU Mapping

The butterfly structure maps naturally to GPU execution:

1. **Stage decomposition.** The $\log_2 N$ butterfly stages execute sequentially; within each stage, all $N/2$ butterfly operations execute in parallel.
2. **Thread mapping.** Each CUDA thread computes one butterfly operation: load two elements, multiply one by a twiddle factor, add/subtract, store.

3. **Shared memory.** For $N \leq 4096$, the entire polynomial fits in shared memory (48 KB at 64-bit coefficients), eliminating global memory round-trips between stages.

4. **Large N.** For $N = 16384$ or $N = 65536$ (CKKS), we use a four-step NTT: decompose into two dimensions, NTT each dimension in shared memory, apply twiddle factors in global memory.

Theorem 2.1 (NTT Throughput)**.** *For degree N and batch size B, the GPU NTT achieves throughput*

$$T = \frac{B \cdot N}{t_{\text{kernel}}}$$

butterfly operations per second. For $N = 1024$, $B = 65536$ on an RTX 4090, $T \approx 2.1 \times 10^{11}$ butterflies/s.

2.3 Modular Arithmetic

All NTT operations use 64-bit modular arithmetic. We implement Barrett reduction for general moduli and Montgomery multiplication for NTT-friendly primes where $q \equiv 1 \pmod{2N}$.

Lemma 2.1 (Montgomery Cost)**.** *Montgomery multiplication on a GPU requires 4 integer multiplications and 2 additions per modular multiply, versus 6 multiplications for Barrett reduction. For NTT-dominated workloads, Montgomery provides a $1.3\times$ throughput improvement.*

3 FHE Bootstrapping on GPU

3.1 TFHE Bootstrap Structure

Each TFHE programmable bootstrap consists of:

1. **Blind rotation.** n external products between an RLWE ciphertext and elements of the bootstrapping key. Each external product decomposes the RLWE ciphertext into ℓ polynomials (gadget decomposition), multiplies each by a bootstrapping key element (via NTT), and accumulates.

2. **Sample extraction.** Extract an LWE ciphertext from the RLWE accumulator. This is a coefficient extraction with no arithmetic cost.

3. **Key switching.** Matrix-vector multiplication to switch from the bootstrapping key to the original key.

3.2 GPU Parallelism

- **Inter-gate parallelism.** Independent gates bootstrap concurrently. A batch of B gates launches B independent blind rotations.

- **Intra-gate parallelism.** Within a single blind rotation, the ℓ gadget-decomposed polynomial multiplications are independent and parallelise across thread blocks.

- **NTT reuse.** The bootstrapping key polynomials are pre-transformed to NTT domain and stored in GPU global memory, eliminating redundant forward NTTs.

3.3 Memory Management

The bootstrapping key for TFHE ($n = 630$, $N = 1024$, $\ell = 3$) occupies $630 \times 3 \times 2 \times 1024 \times 8 \approx 29$ MB in NTT domain. This fits in GPU global memory on all target devices.

For CKKS ($N = 16384$), the evaluation key is larger ($\sim$1.5 GB for full key switching keys). We use a streaming approach: evaluation key chunks are loaded on demand, overlapping computation with memory transfer.

3.4 Benchmarks

Table 1: TFHE bootstrapping on GPU (single gate, $n = 630$, $N = 1024$).

Device	**Single (ms)**	**Batch 1K (µs/gate)**	**Speedup**
CPU (1 core)	8.2	8200	1×
CPU (16 cores)	0.51	510	16×
RTX 3060	0.18	12	683×
RTX 4090	0.09	5.2	1577×
A100	0.07	3.8	2158×

Batch throughput exceeds single-gate throughput by two orders of magnitude due to full GPU occupancy.

4 GPU NTT for Post-Quantum Cryptography

4.1 ML-KEM and ML-DSA

The post-quantum standards adopted by Pars (PNP-004) use polynomial rings $\mathbb{Z}_q[X]/(X^{256} + 1)$ with $q = 3329$ (ML-KEM) and $q = 8380417$ (ML-DSA). Individual NTTs are small ($N = 256$) but validators must process thousands per second during key exchange bursts.

4.2 Batch Key Exchange

During mesh network formation (PNP-002), a node may simultaneously establish sessions with hundreds of peers. Each ML-KEM encapsulation requires 4 NTTs. A batch of 1000 key exchanges requires 4000 NTTs.

Table 2: Batch ML-KEM-768 encapsulation on GPU.

Device	**1K encaps (ms)**	**Throughput (ops/s)**
CPU (1 core)	42.0	23,800
CPU (16 cores)	2.8	357,000
RTX 3060	0.9	1,110,000
RTX 4090	0.4	2,500,000

4.3 Batch Signature Verification

ML-DSA signature verification for block validation requires batch NTT. A block containing 1000 transactions with post-quantum signatures requires ∼8000 NTTs. GPU acceleration keeps verification within the block time budget.

5 Batch ZK Proof Generation

5.1 Groth16 Prover

The ZK identity system (PNP-006) uses Groth16 proofs for credential verification. The Groth16 prover is dominated by two operations:

1. **Multi-scalar exponentiation (MSM).** Computing $\sum_{i=1}^{n} s_i \cdot G_i$ for scalar vector $\mathbf{s}$ and point vector $\mathbf{G}$.
2. **NTT.** Polynomial multiplication for the QAP (Quadratic Arithmetic Program) evaluation.

5.2 GPU MSM

We implement Pippenger's algorithm on GPU:

1. Partition scalars into c-bit windows ($c = 16$ for 256-bit scalars gives 16 windows).
2. For each window, accumulate points into 2^c buckets using atomic additions.
3. Reduce buckets via prefix sum.
4. Combine windows via double-and-add.

Theorem 5.1 (MSM Complexity)**.** *Pippenger MSM over n points with b-bit scalars requires $O(n/\log n)$ group operations. On a GPU with P processors, the parallel complexity is $O(n/(P \log n))$ plus $O(b/c \cdot 2^c)$ for bucket reduction.*

5.3 Batch Identity Proofs

During onboarding events (diaspora community gatherings, cultural events), many users may generate identity proofs simultaneously. The GPU batch prover generates multiple Groth16 proofs in parallel.

Table 3: Groth16 proof generation for ZK-ID credential (2^{16} constraints).

Device	Single (ms)	Batch 100 (ms/proof)
CPU (16 cores)	1850	1850
RTX 3060	420	62
RTX 4090	180	24
A100	120	15

Batch proving on A100 achieves 123× speedup over CPU, enabling community onboarding events where hundreds of identity proofs are generated in minutes.

6 Unified Kernel Library

6.1 Architecture

The Pars GPU library (`pars-gpu`) provides three layers:

1. **Polynomial arithmetic.** NTT, inverse NTT, pointwise multiplication, gadget decomposition. Parameterised over modulus q, degree N, and number of limbs.
2. **Cryptographic primitives.** FHE bootstrap, ML-KEM encapsulation, Groth16 MSM. Built on the polynomial layer.
3. **Application APIs.** Encrypted search, batch key exchange, identity proof generation. Built on the primitive layer.

6.2 Portability

Kernels are written in CUDA with a HIP translation layer for AMD GPUs. A CPU fallback path uses AVX-512 SIMD for x86 and NEON for ARM, ensuring functionality on all diaspora hardware.

6.3 Memory Safety

All GPU memory allocations are:

- Zeroed on free (sensitive key material).
- Bounds-checked via a lightweight allocator wrapper.
- Pinned (page-locked) for DMA transfers to avoid swapping sensitive data to disk.

7 Secure Offloading

Resource-constrained nodes (mobile devices, low-power mesh nodes) may offload GPU-intensive operations to GPU-equipped peers via the mesh network (PNP-002).

7.1 Protocol

1. The requesting node encrypts the computation input under an FHE key (PNP-009) or uses a verifiable computation protocol.
2. The encrypted input is sent to a GPU-equipped peer.
3. The peer performs the GPU-accelerated computation on the encrypted data.
4. The result is returned; the requesting node decrypts or verifies locally.

Theorem 7.1 (Offloading Privacy)**.** *Under the FHE security model (PNP-009), the GPU peer learns nothing about the plaintext inputs or outputs of the offloaded computation.*

For ZK proof generation, a verifiable computation approach is more efficient: the peer generates the proof, and the requesting node verifies it (verification is cheap).

Table 4: GPU speedup summary across Pars cryptographic operations.

Operation	CPU 16c (ms)	RTX 4090 (ms)	Speedup
FHE search (1K msgs)	8200	1400	5.9×
FHE bootstrap (1K batch)	510	5.2	98×
ML-KEM (1K encaps)	2.8	0.4	7×
ML-DSA verify (1K sigs)	4.2	0.6	7×
Groth16 prove (batch 100)	185K	2400	77×

8 Performance Summary

9 Related Work

cuFHE. Dai and Sunar [1] demonstrated GPU TFHE bootstrapping. Pars extends cuFHE with batch scheduling, NTT kernel sharing across FHE/PQC/ZK, and constant-time guarantees.

Icicle. Ingonyama's Icicle [2] provides GPU primitives for ZK proving. Pars adopts a similar MSM strategy but integrates it with the unified NTT layer.

Lux GPU Infrastructure. The Lux network's GPU acceleration for consensus BLS signatures provides the deployment infrastructure. Pars reuses the Lux GPU node provisioning and extends it with FHE/PQC workloads.

10 Conclusion

We have presented a unified GPU acceleration framework for the Pars Network that speeds up FHE bootstrapping, post-quantum cryptographic operations, and zero-knowledge proof generation through a shared NTT kernel library. The framework targets both data-centre and consumer GPUs, enabling diaspora users on commodity hardware to participate in privacy-preserving computation. Secure offloading via the mesh network extends GPU acceleration to resource-constrained mobile devices.

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